

Matched Calories, Matched Responses? Comparing Post-Lunch Metabolic Patterns

Abstract

A nutrition label estimates how much energy a meal contains, but not how quickly that energy becomes available. We tested whether two lunches matched on labeled calories and broad macronutrients, but representing minimally processed and ultra-processed meal patterns, produced similar short-term responses. In a randomized within-subjects crossover study, 48 adults ate each lunch on separate days while glucose was tracked continuously; blood samples, later snack intake, and self-reported hunger provided additional outcomes. The glucose curves initially rose similarly but later diverged, and participants ate more after the ultra-processed meal despite reporting similar hunger. The findings show that meals with similar labeled calories and broad macronutrient totals can still produce different short-term physiological and behavioral responses.

Significance. Calorie labels describe total energy, not the timing of the body's response. This study tests whether two meals with similar labeled calories and macronutrients can still produce different glucose patterns and later eating. The findings suggest that calorie matching is informative but does not make meals physiologically interchangeable.

Key Terms

Continuous glucose monitor (CGM): a wearable sensor that estimates glucose repeatedly over time, producing a trajectory rather than a single reading.

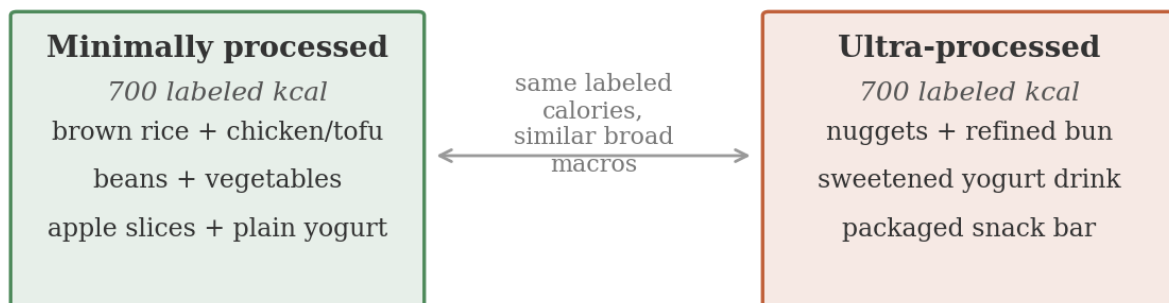
Fasting baseline: a participant's glucose level just before eating the lunch.

Insulin: a hormone that helps move glucose out of the blood and into the body's cells.

Glucose dip: a fall in glucose after its post-meal rise, sometimes below the pre-meal baseline.

Triglycerides: fats carried in the blood; here, measured once after the meal.

Processing level: how far a food has been altered from whole ingredients by industrial preparation (classified here with the NOVA system).



...but the two meals also differ in texture, fiber form, and how fast they are eaten.

Figure 1. The two lunches were matched on labeled calories and broad macronutrient totals and were selected to represent minimally processed and ultra-processed meal patterns.

Introduction

A calorie label estimates how much energy a food contains. It does not indicate how quickly nutrients become available after the food is eaten or how the body's response unfolds over time. On an **energy-equivalence** account, once two meals are matched on calories, carbohydrate, fat, and protein, they should produce broadly similar short-term responses because they provide similar nutritional totals.

A **food-structure** account makes a different prediction. Processing can alter properties such as texture, intact fiber, and whether nutrients are consumed in solid or liquid form. These features may affect how quickly nutrients become available, producing a faster glucose rise and a different later trajectory. Such changes could also be related to later eating, although an association between

glucose and intake would not establish that one caused the other.

The study also distinguishes physiological responses from later eating. Glucose, insulin, and triglycerides describe how the body responds to a meal, whereas snack intake is a behavioral outcome. A meal could change physiological responses without changing later intake, and greater intake could occur for reasons other than glucose dynamics. Earlier work found that people consumed more calories on an ultra-processed diet than on a minimally processed diet matched for broad nutritional content, but that study varied an entire diet over two weeks rather than a single meal (Hall et al., 2019).

We tested whether the two lunches produced different glucose trajectories, whether single-timepoint insulin and triglyceride measurements aligned with those trajectories, whether later snack intake differed, and whether self-

reported hunger showed the same pattern as observed eating.

Method

Design and lunches. 48 healthy adults (54% female; ages 19–55) completed a randomized within-subjects crossover study with two laboratory visits at least one week apart. Each participant ate both lunches, one per visit, with order randomized and counterbalanced. This allowed each person to serve as their own comparison, reducing the influence of stable individual differences such as baseline metabolism and habitual eating patterns. Both lunches provided 700 labeled kilocalories and were matched approximately on protein, fat, and carbohydrate. The minimally processed lunch included brown rice, grilled chicken or tofu, black beans, mixed vegetables, apple slices, and plain yogurt. The ultra-processed lunch included chicken nuggets, a refined-flour bun or crackers, a sweetened yogurt drink, and a packaged snack bar. The meals were selected to represent minimally processed and ultra-processed patterns and were classified using the NOVA system.

Measures. Participants wore a continuous glucose monitor that recorded glucose every 15 minutes across a four-hour observation period, providing a trajectory of the post-meal rise and fall. Insulin and triglycerides were measured from blood samples taken immediately before lunch and once 60 minutes afterward, providing a single post-meal snapshot rather than a full trajectory. Four hours after lunch, participants received a tray of mixed sweet and savory snacks for 30 minutes. The tray was weighed before and after to estimate calorie intake. Immediately before the tray appeared, participants rated how hungry they felt on a 1–7 scale. Glucose responses were summarized by peak level and time spent below the pre-meal baseline. The two conditions were compared within participants, and snack intake was examined in relation to each participant’s lowest post-meal glucose level.

Results

Similar beginning, different later trajectory. For the first half hour, glucose rose at much the same rate after both lunches. The curves then parted. The ultra-processed lunch reached a higher peak (about 142 vs. 116 mg/dL) and, over the next two hours, fell further, spending far longer below the pre-lunch baseline than the minimally processed lunch, which produced a flatter, more sustained curve (Fig. 2). The point is less the exact numbers than the shape: two meals that looked alike early on diverged later. A single well-timed blood draw could have caught the shared early rise and missed this later split entirely, which is why the repeated CGM record carries information a one-off measurement cannot.

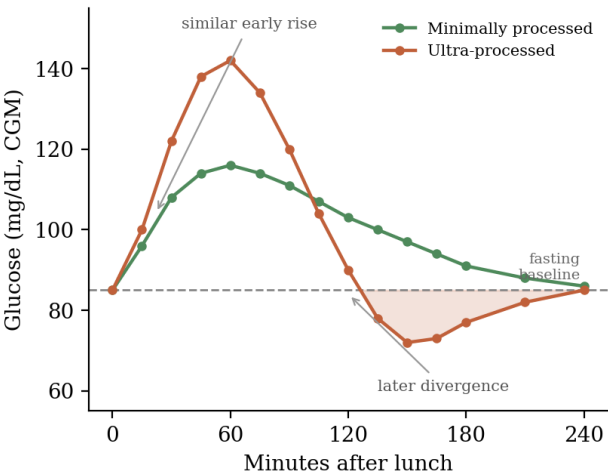


Figure 2. Mean glucose across four hours relative to the pre-meal baseline. The curves initially rise together and then diverge, illustrating how repeated measurement reveals a later difference that a single reading could miss.

Single-timepoint blood measures. Insulin was higher 60 minutes after the ultra-processed lunch, broadly aligning with the glucose pattern. Triglycerides were only modestly higher. Because both outcomes were measured at a single post-meal timepoint, they provide less information about the timing and shape of the response than the repeated glucose measurements.

Measure	Min.	Ultra
Glucose peak (mg/dL)	116	142
Time below baseline (min)	6	38
Insulin at 60 min (μIU/mL)	31	52
Triglycerides at 60 min (mg/dL)	104	118
Afternoon snack (kcal)	186	329
Hunger before tray (1–7)	4.8	5.1

Table 1. Physiological, behavioral, and subjective outcomes by lunch condition. The measures describe different aspects of the post-meal response and need not change together.

Later eating and reported hunger. Participants ate more from the afternoon snack tray after the ultra-processed lunch. However, their hunger ratings immediately before the tray differed little between conditions. This dissociation indicates that self-reported hunger and observed intake were not interchangeable measures of the post-meal response.

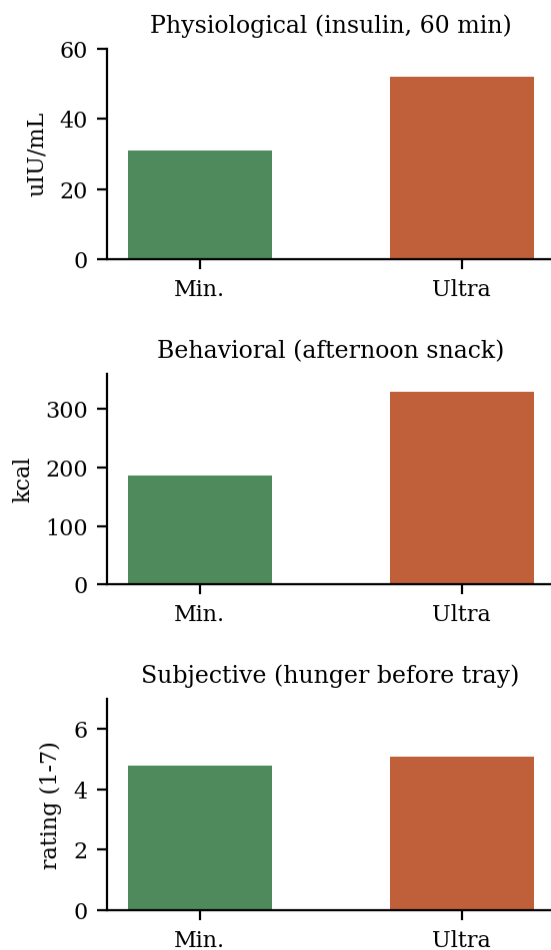


Figure 3. Insulin response, afternoon snack intake, and self-reported hunger by lunch condition. Insulin and snack intake differed between meals, whereas reported hunger was similar.

Association between glucose and later eating.

Participants whose glucose fell further below baseline also tended to eat more from the snack tray. This pattern is consistent with a possible relationship between glucose dynamics and later eating, but the study did not manipulate glucose independently and therefore cannot establish that the decline caused the additional intake.

Conclusion

Two lunches with the same labeled calories and similar broad nutrient totals did not produce the same short-term response. The findings are more consistent with the food-structure account than with simple energy equivalence: matched totals were not enough to produce matched glucose trajectories. The metabolic and behavioral findings should be kept distinct: a different glucose curve and a larger afternoon snack are two outcomes, not one, and their moving together is a suggestive association rather than a demonstrated mechanism. The single insulin and triglyceride draws, likewise, support weaker claims about timing than the repeated glucose record does.

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